

MIT Lab Assignment 24/09/26

Task 1: Projection of a Vector onto Another Vector

Given:

- $a = (4, 2, 0)$
- $b = (1, 2, 2)$

1. Manually compute $\text{proj}_b(a)$.
2. Use Python to verify the result.
3. Compute the perpendicular component:

$$a_{\perp} = a - \text{proj}_b(a)$$

Task 2: Projection onto a Subspace

Given the subspace W spanned by:

$$v_1 = (1, 0, 1), \quad v_2 = (0, 1, 1)$$

and vector:

$$x = (2, 3, 4)$$

1. Orthonormalize v_1, v_2 using Gram–Schmidt.
2. Construct projection matrix $P = QQ^T$ where Q is the orthonormal basis matrix.
3. Compute the projection Px .
4. Verify using NumPy.

Task 3: Application – Image Compression with SVD

1. Load a grayscale image (use “anyimage.png” or any 256×256 image).
2. Perform SVD on the image matrix.
3. Reconstruct the image using only the first $k = 20, 50, 100$ singular values.
4. Display the original and reconstructed images.
5. Comment on:
 - Quality difference
 - Storage reduction
 - Role of singular values